

Midterm Project Template File

Module 5

Graded at 100 points

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	Analysis to be performed:	EXCEL COMMAND USED (if applicable):	NUMERIC RESULT or text-based response:
Section 2. DESCRIPTIVE STATISTICS (20 points total, 2 points each)	a) Mean		
	b) Median		
	c) Standard Deviation		
	d) Variance		
	e) Minimum		
	f) Maximum		
	g) 1 st Quartile		
	h) 3 rd Quartile		
	i) Compare the mean on-time arrival results for AIRLINE #1 vs. AIRLINE #2 and state a conclusion.		
	j) Compare the median on-time arrival results for AIRLINE #1 vs. AIRLINE #2 and state a conclusion.		

Section 3. TRIMMED MEAN (10 points, 2.5 points each)	a) Calculate a 20% trimmed mean.		
	b) Does this trimmed mean represent a significantly different result as compared to the untrimmed mean calculated in Step 2a?		
	c) Explain why the results were significantly different or explain why the results were not significantly different.		
	d) Why would someone want to calculate a trimmed mean instead of using the untrimmed mean?		
Section 4. SIDE-BY-SIDE BOXPLOTS (20 points total, 4 points each)	a) Copy and paste the side-by-side boxplot here:		
	b) Examine each boxplot separately, and comment on the shape of the distribution with respect to symmetric or skewed for each airline.		

	c) Are there any outliers shown on the boxplots? If there are any outliers shown on the boxplot, comment on what this indicates with respect to on-time arrivals. Should the airline be concerned?		
	d) Examining the boxplots, do you observe any significant differences in the distribution of on-time arrivals data for the two airlines?		
	e) Assume you are a manager at AIRLINE #1. State a conclusion based on your analysis of the side-by-side boxplots.		
Section 5. PARETO CHART (10 points, 2 points each)	a) Copy and paste the Pareto chart:		

	b) Based on the Pareto chart, which is the worst performing airline for average on-time arrivals?		
	c) Based on the Pareto chart, which is the best performing airline for average on-time arrivals?		
	d) What is the benefit of creating a Pareto chart versus a bar chart?		
	e) Assume you are a manager at the airline with the worst performance. What steps would you investigate to improve on-time arrival? What factors might be within the control of the airline and which factors might be out of the control of the airline with respect to on-time arrival performance?		
Section 6. APPLYING NORMAL DISTRIBUTION (20 points, 3.33 points each)	a) Calculate the probability that a flight arrives early. Include the Excel command used.		
	b) Calculate the probability that a flight arrives late. Include the Excel command used.		

	c) Calculate the probability that the on-time arrival of a flight falls between the 1 st quartile and 3 rd quartile as determined from Steps 2g and 2h. Include the Excel command used.		
	d) Is the result from part (c) reasonable? What is the expected area under the normal curve between the 1 st and 3 rd quartile? How does your actual numeric result from part (c) compare with the expected area?		
	e) Calculate the probability that the on-time arrival of a flight falls between the 2 standard deviations below the mean and 2 standard deviations above the mean. Include the Excel command used.		
	f) Is the result from part (e) reasonable? What is the expected area under the normal curve between 2 standard deviations below the mean and 2 standard deviations above the mean? How does your actual numeric result from part (e) compare with the expected area?		
Section 7. INVERSE NORMAL DISTRIBUTION (10 points, 5 points each)	a) What on-time arrival delay cuts off the top 10% of the distribution? Include the Excel command used.		

	b) What on-time arrival delay cuts off the bottom 15% of the distribution? Include the Excel command used.		
Section 8. CHEBYSHEV'S THEOREM (10 points, 3.3 points each)	a) For this specific question, we will assume that the distribution of age of aircraft is skewed left. Using Chebyshev's Theorem, find the probability that the age of aircraft falls within 3 standard deviations of the mean. Show details of your calculation.		
	b) Using the normal distribution, find the probability that the age of aircraft falls within 3 standard deviations of the mean. Show details of your calculation and include any Excel commands used.		
	c) Compare your results from parts (a) and (b). What do you conclude about the use of Chebyshev's Theorem?		